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① $X[z] = \dfrac{3 + 2z^{-1}}{1 - 3z^{-1} + 2z^{-2}}$

a) long division

$$
\begin{array}{r|l}
3 + 11z^{-1} + 27z^{-2} & \\
\hline
1 - 3z^{-1} + 2z^{-2} & 3 + 2z^{-1} \\
3 - 9z^{-1} + 6z^{-2} & \\
\hline
11z^{-1} - 6z^{-2} & \\
11z^{-1} - 33z^{-2} + 22z^{-3} & \\
\hline
27z^{-2} - 22z^{-3} & \\
27z^{-2} - 81z^{-3} + 54z^{-4} & \\
\hline
59z^{-3} - 54z^{-4} & .
\end{array}
$$

$X[z] = 3 + 11z^{-1} + 27z^{-2}$

$x[n] = [\,3 \;\; 11 \;\; 27\,]$

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b). Partial fraction.

$$X[z] = \frac{3 + 2z^{-1}}{1 - 3z^{-1} + 2z^{-2}} = \frac{3 + 2z^{-1}}{\underbrace{(-2z^{-1} + 1)}_{A}\underbrace{(-z^{-1} + 1)}_{B}} = \frac{A}{(-2z^{-1} + 1)} + \frac{B}{(-z^{-1} + 1)}.$$

mencari A.

$$A = X[z].(-2z^{-1} + 1) = \frac{3 + 2z^{-1}}{(-2z^{-1} + 1)(-z^{-1} + 1)} \times (-2z^{-1} + 1)$$

$-2z^{-1} + 1 = 0$

$-2z^{-1} = -1$

$z^{-1} = \frac{-1}{-2} = \frac{1}{2}.$

$$= \frac{3 + 2(\frac{1}{2})}{(-\frac{1}{2}) + 1} = \frac{3 + 1}{\frac{1}{2}} = \frac{4}{1/2} = 8$$

mencari B.

$$B = X[z].(-z^{-1} + 1).$$

$$= \frac{3 + 2z^{-1}}{(-2z^{-1} + 1)(-z^{-1} + 1)} \times (-z^{-1} + 1)$$

$-z^{-1} + 1 = 0$

$-z^{-1} = -1$

$z^{-1} = 1.$

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$$= \frac{3 + 2(1)}{-2(1) + 1} = \frac{3 + 2}{-2 + 1} = \frac{5}{-1} = -5$$

$$X[z] = 8 \cdot \frac{1}{(1 - 2z^{-1})} - 5 \cdot \frac{1}{(1 - z^{-1})}$$

$$x[n] = 8\,(2)^n u[n] - 5\,u[n].$$

SIDU

[5]. $x[n] = 8 (2)^n u[n] - 5 u[n]$.

$n = 0$

$x[0] = 8 (2)^0 u(0) - 5 u(0)$

$= 8 \cdot 1 \cdot 1 - 5 \cdot 1 = ?$

$n = 1$

$x[1] = 8 (2)^1 u(1) - 5 u(1)$

$= 8 \cdot 2 \cdot 2 - 5 \cdot 1$

$= 27$.

$n = 2$.

$x[2] = 8 (2)^2 u(2) - 5 u(2)$

$= 8 \cdot 4 \cdot 4 - 5 \cdot 1$

$= 123$.

[2]. $X[z] = \dfrac{1 + 4z^{-1} + 6z^{-2}}{1 + 4z^{-1} + 3z^{-2}}$

$3z^{-2} + 4z^{-1} + 1 \;\big|\; 6z^{-2} + 4z^{-1} + 1$ (quotient 2)

$6z^{-2} + 8z^{-1} + 2$

$0 \quad -4z^{-2} + 2 - 1$

$X[z] = 2 + \dfrac{-4z^{-2} + 2}{3z^{-2} + 4z^{-1} + 1}$.

$F[z] = 2 + \dfrac{2 - 4z^{-2}}{1 + 4z^{-1} + 3z^{-2}}$

$= \dfrac{A}{(3z^{-1} + 1)} + \dfrac{B}{(z^{-1} + 1)}$

• mencari nilai A.

$A = \dfrac{2 - 4z^{-2}}{(3z^{-1} + 1)(z^{-1} + 1)}$ $\quad (3z^{-1} + 1) \;\big|\; 3z^{-1} + 1 = 0$, $3z^{-1} = -1$, $z^{-1} = -\frac{1}{3}$

$= \dfrac{2 - 4(-\frac{1}{3})^{-2}}{(-\frac{1}{3}) + 1}$.

$= \dfrac{2 - 4(9)}{-\frac{2}{3}} = \dfrac{2 - 36}{-\frac{2}{3}} = \dfrac{-34}{-\frac{2}{3}} = 51$

mencari nilai B

$B = \dfrac{2 - 4z^{-2}}{(3z^{-1} + 1)(z^{-1} + 1)}$ $\quad (z^{-1} + 1) \;\big|\; z^{-1} + 1 = 0$, $z^{-1} = -1$

$= \dfrac{2 - 4(-1)^{-2}}{3(-1) + 1}$

$= \dfrac{2 - 4(1)}{-3 + 1} = \dfrac{-2}{-2} = 1$.

$F[z] = 51 \dfrac{1}{1 + 3z^{-1}} + 1 \dfrac{1}{1 + z^{-1}}$

$= 51 (-3)^n u[n] + (-1)^n u[n]$.

$x[n] = 2\delta(n) + 51(-3)^n u(n) + (-1)^n u[n]$

$x[0] = 2\delta(0) + 51(-3)^0 u(0) + (-1)^0 u(0)$

$= 2 \cdot 1 + 51 \cdot 1 \cdot 1 + 1 \cdot 1$

$= 2 + 51 + 1$

$= 54$

$x[1] = 2\delta(1) + 51(-3)^1 u(1) + (-1)^1 u(1)$

$= 2 \cdot 0 + 51(-3)(-3) + (-1)(-1)$

$= 459 + 1$

$= 500$

$x[2] = 2\delta(2) + 51(-3)^2 u(2) + (-1)^2 u(2)$

$= 2 \cdot 0 + 51 \cdot 9 \cdot 9 + 1 + 1$.

$= 4131$.

3. $X[z] = \dfrac{z^3 + 5z^2 + 6z}{z^3 + 2z^2 - 5z - 6}$.

$X[z] = \dfrac{z^3 + 5z^2 + 6z}{z^3 + 2z^2 - 5z - 6} = \dfrac{z(z^2 + 5z + 6)}{z^3 + 2z^2 - 5z - 6} \rightarrow (z+2)(z+3) = 0$

$\dfrac{X[z]}{[z]} = \dfrac{z^2 + 5z + 6}{(z+1)(z-2)(z+3)} = \dfrac{A}{(z+1)} + \dfrac{B}{(z-2)} + \dfrac{C}{(z+3)}$ ✓

$\dfrac{z+2}{(z+1)(z-2)}$

$\dfrac{A}{z+1} + \dfrac{B}{z-2}$

• mencari nilai A.

$A = \dfrac{z^2 + 5z + 6}{(z+1)(z-2)(z+3)}\ (z+1)$ | $z+1=0$, $z=-1$

$A = \dfrac{(-1)^2 + 5(-1) + 6}{(-1-2)(-1+3)}$

$= \dfrac{1 - 5 + 6}{(-3)(2)} = \dfrac{2}{-6} = -\dfrac{1}{3}$ ✓

$A = \left.\dfrac{z+2}{z-2}\right|_{z=-1}$

$= \dfrac{1}{-3} = -\dfrac{1}{3}$

$-1+1=0$

$-1=-1$

• mencari nilai B.

$B = \dfrac{z^2 + 5z + 6}{(z+1)(z-2)(z+3)}\ (z-2)$ | $z-2=0$, $z=2$.

$= \dfrac{(2)^2 + 5(2) + 6}{(2+1)\quad(2+3)}$

$= \dfrac{4 + 10 + 6}{(3)(5)} = \dfrac{20}{10} = 2$ ✗

3

$B = \left.\dfrac{z+2}{z+1}\right|_{z=2}$

$= \dfrac{4}{3}$.

• mencari nilai C.

$C = \dfrac{z^2 + 5z + 6}{(z+1)(z-2)(z+3)}\ (z+3)$ | $z+3=0$, $z=-3$.

$= \dfrac{(-3)^2 + 5(-3) + 6}{(-3+1)(-3-2)(}$

$= \dfrac{9 - 15 + 6}{(-3)(-6)}$ ✓

$= \dfrac{0}{18} = 0$.

$\dfrac{X[z]}{[z]} = \dfrac{-\frac{1}{3}z}{(z+1)} + \dfrac{2z}{(z-2)} + \dfrac{0z}{(z+3)}$. ✓ ✗

$x[n] = -\dfrac{1}{3}(-1)^n u[n] + 2(2)^n u[n] + (-1)^n u[n]$ ✗ ✓